Methodological Basis for the Power–Interest Matrix

1. Multidimensional Assessment Scale

This assessment scale is grounded in stakeholder mapping and adopts Mendelow's power–interest matrix as its principal classification framework. For the power dimension, it draws on resource dependence theory and strategic contingency theory to assess stakeholders' capacity to shape project outcomes across five subdimensions: decision-making authority, control over resources, informational advantage, organizational hierarchy, and social influence. For the interest dimension, it integrates the urgency attribute of stakeholder salience theory with the risk–return analytical logic of strategic management. Five indicators—financial gains and losses, strategic alignment, risk exposure, frequency of attention to relevant issues, and sunk investment costs—are used to assess the extent of stakeholders' vested interest in project outcomes. The scale thus provides a theoretically grounded, operational assessment instrument for the systematic identification and classification of stakeholders and the development of differentiated stakeholder communication strategies.

Power Dimension

Assessment Item	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Decision-making authority					
Resource control					
Informational advantage					
Organizational hierarchy					
Social influence					

Interest Dimension

Assessment Item	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Financial gains and losses					
Strategic alignment					
Risk exposure					
Frequency of issue-related attention					
Sunk investment costs					

2. Expert Assessment

Using a triangulation approach, the assessment involved 12 project team members, six public health experts, six medical experts, and 12 target users (including hospital patients, participants associated with nursing homes and kindergartens, and members of the general public). Ratings were obtained through a Delphi expert consultation process.

Round 1: Participants rated the stakeholders independently, after which the group medians were calculated.

Stakeholder	Power Median	Power IQR	Interest Median	Interest IQR
Policymakers	23	3	10	4
Public health agencies	22	2	24	2
Healthcare institutions	17	3	21	2
Nursing homes and kindergartens	20	3	24	2
Educational institutions	6	1	5	2
Regulatory authorities	21	2	9	3
Patients and high-risk populations	18	2	23	2
Corporate partners	13	3	17	3
Researchers in related fields	19	2	8	2

Round 2: The group medians and each participant's previous ratings were fed back to the

experts, who then revised their ratings to facilitate convergence.

Stakeholder	Power Median	Power IQR	Interest Median	Interest IQR
Policymakers	24	1	11	1
Public health agencies	23	1	25	0.5
Healthcare institutions	18	1	22	1
Nursing homes and kindergartens	19	1	25	0.5
Educational institutions	5	0.5	6	1
Regulatory authorities	21	1	9	1
Patients and high-risk populations	17	1	24	1
Corporate partners	12	1	20	1
Researchers in related fields	16	1	8	0.5

All interquartile ranges (IQRs) in Round 2 were ≤ 1. Accordingly, the Round 2 medians were taken as the final scores and used to construct the scatter plot.

3. References

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